

Total no. of Pages:

Roll. No:

SUMMER SEMESTER, B. TECH (ECE / EE/ICE/EIOT)

END-SEMESTER EXAMINATION, July 2024

Course Code: ECECC02/ EEECC02/ICECC02/ EIECC02

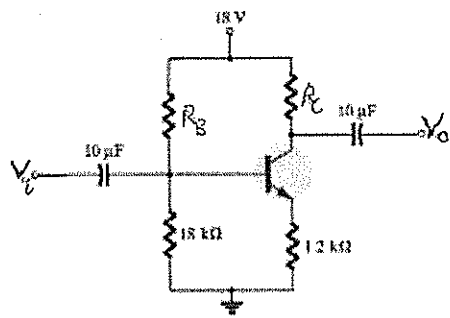
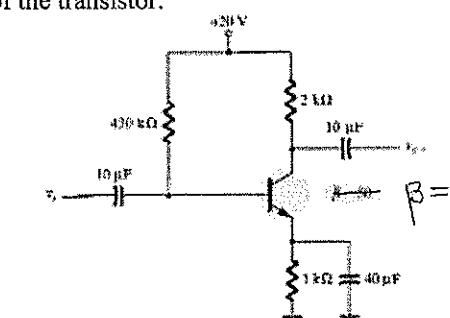
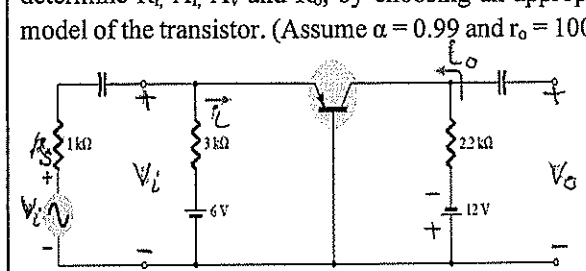
Course Title: Electronic Devices and Circuits

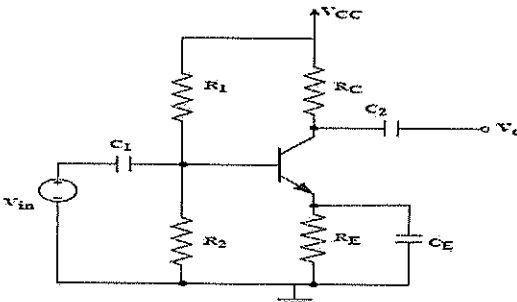
Time: 3:00 Hours

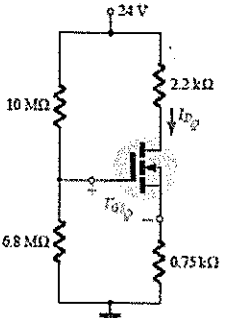
Max. Marks: 40

Note: Missing data/information, if any, may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Attempt any two parts		
(a)	A specimen of metal has square cross section area of $3\text{mm} \times 3\text{mm}$ and its length is 5cm . A potential difference of 1V applied across its length, gives a current of 6mA . Calculate concentration of electron and drift velocity (assume mobility of electron = $1300\text{cm}^2/\text{V-s}$)	4	CO1
(b)	Explain the following terms briefly: a. Doping and Diffusion Length b. Law of Mass Action and Carrier Mobility	4	CO1
(c)	Explain and derive the expression of the barrier potential of PN junction diode.	4	CO1
2.	Attempt any two parts		
(a)	Draw the input, output and transfer characteristics of a BJT and explain how the small signal parameters r_{π} , r_e , g_m and r_o (where the symbols have their usual meaning) are derived from these.	4	CO2
(b)	Determine the values of the resistors R_B and R_C , assuming that β of the transistors is large enough to assume $I_E = I_C$.	4	CO2

			
(c)	For the circuit shown below, determine the operating point of the transistor.	4	CO2
			
3.	Attempt any two parts		
(a)	For the common-base amplifier shown in Figure below, determine R_i , A_i , A_v and R_o , by choosing an appropriate model of the transistor. (Assume $\alpha = 0.99$ and $r_o = 100\text{k}\Omega$)	4	CO3
			

(b)	Using appropriate analysis explain, the circuit of a simple current mirror circuit using BJTs.	4	CO3
(c)	For the common emitter amplifier circuit given below, determine the expressions for (i) R_{in} (ii) R_{out} (iii) A_v	4	CO3
			
4.	Attempt any two parts		
(a)	For an NMOS source follower circuit derive the expressions for its voltage gain (A_v), input resistance (R_{in}) and output resistance (R_o).	4	CO4
(b)	For the NMOS transistor circuit shown in Figure below, determine: (a) I_{DQ} and V_{GSQ} . (b) V_{DS} and V_D Assume $V_{th} = 3V$ and $K = 50mA/V^2$	4	CO4

			
(c)	With suitable mathematical analysis, explain the function of a JFET or MOSFET as an approximately linear voltage-controlled resistance (VCR).	4	CO4
5.	Attempt any two parts		
(a)	Explain the basic principle and working of photodiode.	4	CO5
(b)	Draw and explain the V-I characteristics of Uni-Junction Transistor.	4	CO5
(c)	Explain any two applications of Light Emitting Diodes with appropriate circuit diagrams.	4	CO5